Al-Balqa Applied University

Faculty of Engineering Technology, Scientific Basic Science Department, MID Exam

Linear Algebra.

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	A	B	C	C	D	A	B	C	D	B

Question One: The system $x + y = \pi$, $2x+y=2\pi$ **has**

A) a unique solution B) infinitely many solutions C) no solution
D) two solutions E)None

Question Two: If the system $x+2y=a$, $3x+6y=a^2$ **has infinitely many solutions, then the value of the positive constant** a **equals**

A) 2 B) 3 C) 4 D) 5 E) None

Question Three: For a nonsingular matrix A **with**

$$A-1=\begin{bmatrix} 1 & 1 \\ 0.5 & 1.5 \end{bmatrix}$$

Then the solution of the system $A^{-1}X=[]$ **is** $\begin{matrix} 0 \\ 2 \end{matrix}$

A) $(-3,-2)$ B) $(2,-2)$ C) $(-2,2)$ D) $(-2,3)$
E) None

Question Four: Let I **be the** 3×3 **identity matrix, then**

$$((trace(I)\, I)^{-1})t=$$

A) $\frac{1}{5}I$ **B)** $\frac{1}{4}I$ **C)** $\frac{1}{t_3}$ **D)** $\frac{2}{9}I$ **E) None**

Question Five: Considera $n \times n$ **matrix** A **with** $det(A) = 2$ **. If** $det(nA^{-1}) = \frac{27}{2}$ **, then** $n=$

A) 5 B) 2 C) 4 D) 3 E) None

Question Six: Consider the vector space $V=\mathbb{R}^3$ **where**

$$(x_1, y1, z_1) \oplus (x_2, y_2, z2) = (x_1 + x_2 - 1, y_1 + y_2, z_1 + z_2 - 1)$$

Then the zero vector $OV=$

A) (1,0,1) B) (1,1,0) C) (1,0,0)
D) (0,0,0) E) None

Question Seven: The constants that make $p=x^2-6x$ **a linear combinations of** $p_1= x^2$ **and** $p_2=3x$ **are**

A) 1,−3 B) 1,−2 C) 1,3 D) 1,2 E) None

Question Eight: For the matrix

$$A = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$$

We have $det(A^{-1}) + trace(A^{-1}) =$

A) 3 B) 2 C) 14 D) 15 E) None

Question Nine: If the matrix

$$A = \begin{bmatrix} 0.5 & a-4 & 0 \\ 1 & 0.5 & 1 \\ 0 & 1 & 0.5 \end{bmatrix}$$

is symmetric, find the value of the constant a

A) 2 B) 3 C) 4 D) 5 E) None

Question Ten: If $det\left(\begin{bmatrix} a & b \\ 1 & 2 \end{bmatrix}\right) = 5$ **; then** $det\left(\begin{bmatrix} 2 & 4 \\ -a & -b \end{bmatrix}\right) =$

A) 20 B) 10 C)−10 D) 5 E)None

Question Eleven: Find ; if any , the solution(s) of the system
x+y=3 , 2x+y=5 using Gauss-Jordan Elimination

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -1 \end{array}\right]$$

$$R_2 = -R_2 \longrightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array}\right]$$

$$R_1 = R_1 - R_2 \quad \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right]$$

Solution: $(x, y) = (2, 1)$

Question Twelve: Show that $W=\{(a,b) \in R^2$ (a) is the system
R2 under the natural vector addition and scalar multipiĉadization

Proof:

1. Check Closure under addition: Let u=(a, b), v=(c, d) be in W.
We need u+v= (a+c, b+d) in W.

Since u, v are in W, a=b+1 and c=d+1.

Check if u+v satisfies the condition: a+c = (b+1) + (d+1) = b+d+2.

For u+v to be in W, it must satisfy a+c = (b+d)+1.

Since b+d+2 ≠ b+d+1, W is not closed under vector addition.

Thus, W is not a subspace of R2. (Counterexample: (1, 0) and (2, 1) are in W, but their sum (3, 1) is not, since 3 ≠ 1+1).